

Aim: Interface and Package.

1.Create a java package to illustrate the concept of package and demonstrate its usage.

CODE:

STEP 1: Create an folder.

STEP 2: Create an Class (A and B) in that folder.

STEP 3: For class first give-> package packagename;

STEP 4: For class A.

```
package userpack;
public class B{
    public void display(){
        System.out.println("Hello from Package! with class B.");
    }
}
```

STEP 5: For class B.

```
package userpack;
public class A{
    public void display(){
        System.out.println("Hello from Package! with class A.");
    }
}
```

STEP 6: Create a java file that will import the package.

STEP 6: For the main function.

```
import userpack.*;
public class UserPackDemo {
    public static void main(String args[]){
        A a = new A();
        B b = new B();
        a.display();
        b.display();
    }
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\Desktop\Java Programs\userpack>javac *.java
C:\Users\ROHIT\OneDrive\Desktop\Java Programs\userpack>cd..
C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac UserPackDemo.java
C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java UserPackDemo
Hello from Package! with class A.
Hello from Package! with class B.
```

2.Write a java program to illustrate the concept of interface in a java.

CODE:

STEP 1: Create an interface in one file.

```
interface Callback{
    public void callme(int x);
    final int max=100;
}
```

STEP 2: Create an class that can implement that interface (Callback).

```
class InferCall implements Callback{
    public void callme(int x){
```

```
        System.out.println("Square:" + (x*x));
        System.out.println("Value of max:" + max);
    }
    void myFun(){
        System.out.println("Typing.....");
    }
}
public class InterfaceDemo {
    public static void main(String args[]){
        InferCall i = new InferCall();
        i.callme(45);
        i.myFun();
    }
}
```

OUTPUT:

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac InterfaceDemo.java

C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java InterfaceDemo

Square:2025

Value of max:100

Typing.....

3.Create a java package to illustrate the concept of interface and implement in a java program.**CODE:****STEP 1: Create a pack and create and implement the interface.**

```
package userpack;
interface In{
    public int factorial(int i);
}
public class I implements In{
    public int factorial(int i){
        int n = 1;
        for(int j=1;j<=i;j++){
            n = n*j;
        }
        return n;
    }
}
```

STEP 2: Import the package and execute it.

```
import userpack.I;
public class Fact{
    public static void main(String args[]){
        int r;
        I i = new I();
        r = i.factorial(5);
        System.out.println("Factorial of 5 is " + r);
    }
}
```

OUTPUT:

```
C:\Users\ROHIT\OneDrive\Desktop\Java Programs>cd userpack
C:\Users\ROHIT\OneDrive\Desktop\Java Programs\userpack>javac I.java
C:\Users\ROHIT\OneDrive\Desktop\Java Programs\userpack>cd..
C:\Users\ROHIT\OneDrive\Desktop\Java Programs>javac Fact.java
C:\Users\ROHIT\OneDrive\Desktop\Java Programs>java Fact
Factorial of 5 is 120
```